higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

APPLIED ENGINEERING TECHNOLOGY
NQF LEVEL 4

27 FEBRUARY 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 1.1.1 *Kinetic energy* is the energy that a body possesses✓ by virtue of its motion. ✓*Potential energy* is the energy that a body possesses✓ by virtue of its position or state of strain.✓ (4)

1.1.2 *Centrifugal force* is the force that is exerted outward✓ when an object spins or rotates✓. (2)

1.1.3 When a rotational force is applied about a point, torque is generated✓. When the length of the lever changes, while applying the same force, the torque will change✓. A short level will generate less torque✓ while a longer lever generates more torque about the point.✓ (4)

1.2
- Bevel gear
- Spur gear
- Helical gear
- Worm gear
- Double helical gear (5)

1.3 It is a pressure storage reservoir in which a non-compressible hydraulic fluid is held ✓under pressure applied by an external source.✓ (2)

1.4 1.4.1 $W = mg$
$= 80 \times 9{,}81$
$= 784{,}8\ N$✓

$W = NR$

$\mu = \frac{F}{NR}$✓

$= \frac{180}{784{,}8}$✓
$= 0{,}23$✓ (4)

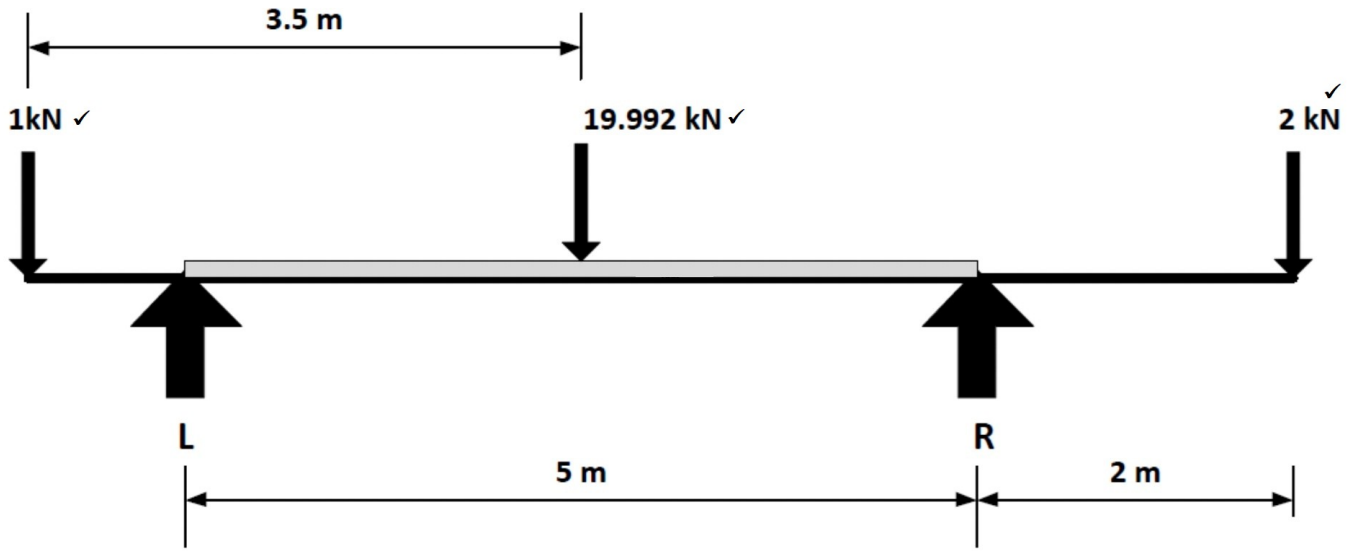
1.4.2 $\tan \theta = \mu$
$\theta = \tan^{-1} \mu$✓
$= \tan^{-1} 0{,}23$✓
$= 13^0$✓ (3)

1.4.3 $F = \mu \times NR$✓
$= 0{,}23 \times (2 \times 784{,}8)$✓✓
$= 361\ N$✓ (4)

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QUESTION 2

2.1



(3)

2.2 REACTIONS:

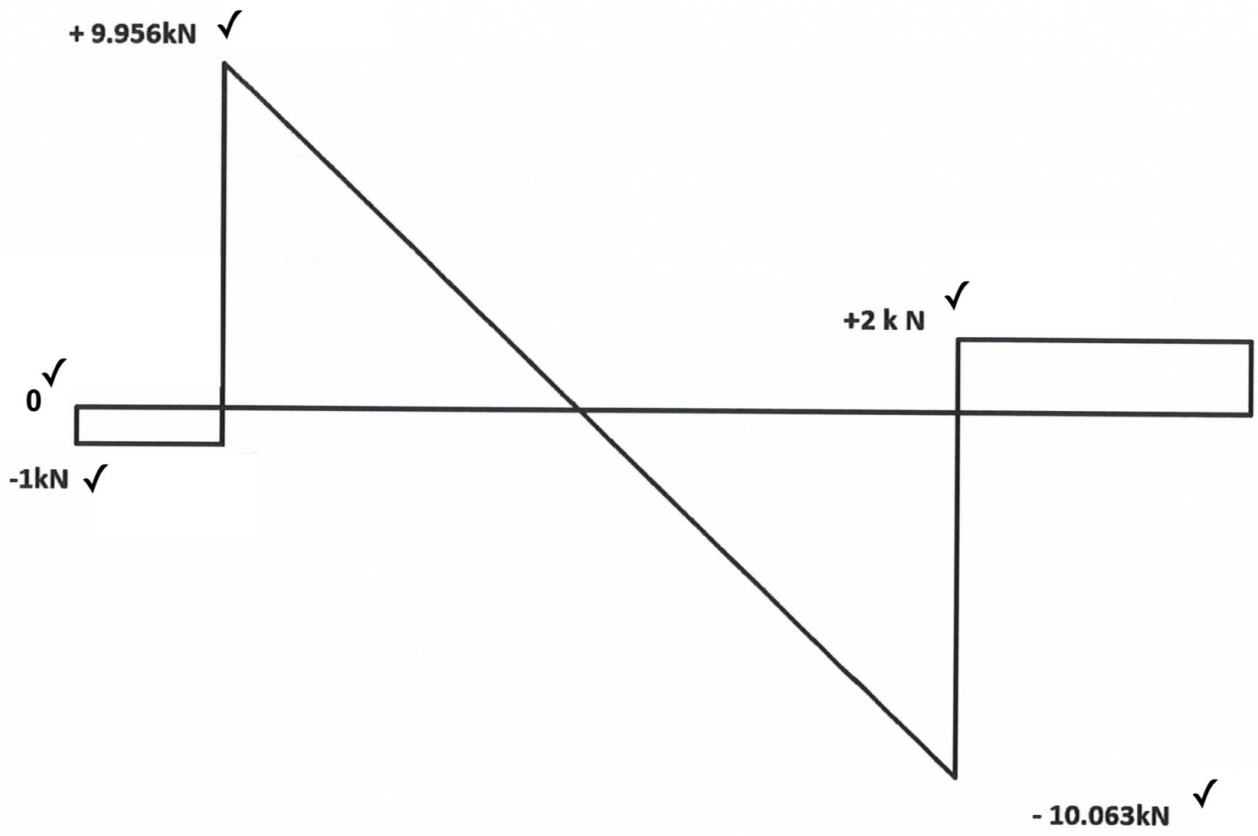
Sum of anti-clockwise moments = Sum of clockwise moments
R × 5 + 1 × 1 + 1,5 × 0,2 = 20 × 2,5 + 2 × 7✓
5R +1 + 0,3 = 50 + 14
5R = 64 – 1,3✓
R = 12,54 kN✓

⇑F = ⇓F
L + R = 1 + 1,5 + 20 + 2✓
L = 24,5 – R✓
L = 24,5 – 12,54
L = 11,96 kN✓

TEST:

Upward forces = Downward forces
12,54 + 11,96 = 1 + 1,5 + 20 + 2
24,5 kN = 24,5 kN✓ (7)

2.3



(5)

2.4 $BM_A = 0$✓

$BM_B = (-1\ 000 \times 0,8) = -800$ Nm✓

$BM_C = (-1\ 000 \times 1) – (1\ 500 \times 0,2) = -1\ 300$ Nm✓

$BM_D = (-1\ 000 \times 6)\ (1\ 500 \times 5,2) + (11\ 960 \times 5) – (2\ 001,24 \times 25)$ ✓
$= -4\ 031$ Nm

$BM_E = 0$✓ (5)

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QUESTION 3

3.1 3.1.1 Computer-aided design software is used by architects, engineers and manufacturers✓ to create precision drawings.✓ (2)

3.1.2
- Open-loop control systems✓
- Closed-loop control systems✓ (2)

3.1.3
- Poor design increases the cost of the project.
- It makes a bad impression on the client.
- Poor design results in a bad product and maybe failure.
- Could bring the company into bad repute.
- The company may not get another contract.
- It may cause delays. (Any 4 × 1) (4)

3.2 Pressure = density × g × h

$$h = \frac{\text{pressure}}{\text{density} \times g}\checkmark$$

$$= \frac{350 \times 10^3\checkmark}{1\,000 \times 9{,}81\checkmark}$$

$$= \frac{350\,000\checkmark}{9810\checkmark}$$

$$= 35{,}678 \text{ m}\checkmark$$ (6)

3.3
- Point of operation where work such as cutting, shaping is performed
- Where power is transmitted to the machine
- Where parts of the machine are moving while the machine is working
- Spillages of for example oil on the floor
- Poor housekeeping (Any relevant answers 5 × 1) (5)

3.4 3.4.1 It forces the oil out of the oil sump

3.4.2 It removes all dirt and impurities from the oil.

3.4.3 Oil is collected in an oil sump from where it is sucked through the system.

3.4.4 It is used to check the oil level.

(4 × 1) (4)

3.5
- Tools lying around
- Open lids on lubricating fluids/solvents
- Waste material used for cleaning equipment/parts lying around
- Components not properly stacked
- Oil/Water/Lubricant spills on floor (or any other applicable hazard)

(Any 2 × 1) (2)
[25]

QUESTION 4

4.1 $3\ 000\ mm^2 = 3\ 000 \times 10^{-6}\ m^2$✓
0,007 kN = (0,007 × 1 000) N✓

$$Pressure = \frac{F}{A}$$

$$= \frac{0{,}007 \times 1\ 000}{3\ 000 \times 10^{-6}}$$✓

= 2333,33 Pa✓
= 2,3 kPa✓ (5)

4.2 4.2.1 D
4.2.2 H
4.2.3 G
4.2.4 B
4.2.5 J
4.2.6 C
4.2.7 A
4.2.8 I
4.2.9 E
4.2.10 F

(10 × 1) (10)
[15]

QUESTION 5

5.1
- Linear
- Rotary
- Reciprocating
- Oscillating (4)

5.2
- The motion of a servo motor is much smoother than that of a stepper motor.
- It is small and extremely powerful for its size.
- The amount of power applied to the motor is proportional to the distance it needs to travel.
- It has feedback mechanisms to sense its position. (4)

5.3
- Open: sewerage and sand
- Semi-open: fibrous materials and sediment
- Closed/Shrouded: clean liquids (3)

5.4 It uses the cooling action of the vaporising refrigerant to cool the inside of a fridge. (1)

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TOTAL: 100

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